LINEAR ORDERS OF EXACT SCOTT SENTENCE COMPLEXITY SIGMA-IN_(LAMBDA+3)

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Editorial note. All ordinals in this Scott-analysis problem are countable, and limit ordinal means nonzero limit. L times Q replaces each rational point with a copy of L, as defined in1017-1020.

Editorial note. The author proof at1556 calls z the second-copy initial element, while the displayed sec(x,z) formula and its subsequent use1557 specify x as that element and z as the initial element of the first copy. The formula is preserved; this variable-description slip is explicitly recorded for independent review.

Editorial note. The upper-bound passage uses the standard named-parameter Scott-rank criterion stated among the inputs. The lower-bound comparison is with A+B; it is not isomorphic to A+L2+B because the latter has a least doubled-ordinal block and the former does not.

Editorial note. Native mathematical macros and theorem declarations are selected from the author dino.sty. Its unrelated draft, bibliography-backend and page-layout options are not loaded. Bibliographic metadata from the supplied flattened.bbl is formatted as a standard bibliography; author mathematical spans are unchanged.

Editorial scope. One exact Scott-sentence complexity construction. The complete bespoke shuffle-sum comparison and author main proof are solution obligations. Classical well-order Scott analysis and the general orbit-to-Scott-sentence criterion are explicit inputs. The previously published Z-power lemma is not re-counted or required.

Allowed standard prerequisites (editorial).

- Infinitary Scott analysis and back-and-forth preservation. A Sigma-in_alpha formula is an infinitary countable disjunction of existential blocks over lower-level Pi formulas, with the dual definition for Pi. The back-and-forth preorder matches those formula classes. If all tuple orbits are Sigma-in_alpha definable over a fixed finite tuple, a Sigma-in_(alpha+2) Scott sentence exists. A nonisomorphic K with L<=_beta K rules out a Pi-in_beta Scott sentence of L.. Source45-86 and the stated standard background173-184.. Translate the actual orbit upper bounds and constructed nonisomorphic back-and-forth witness into exact sentence complexity.
- Classical Scott analysis of well orders and finite condensation. For a nonzero countable limit ordinal lambda, omega^lambda <=_(lambda+1) omega^lambda*2. For each gamma<omega^lambda its interval isomorphism type has infinitary complexity below lambda. The finite-condensation relations start with equality, take finite distance in the quotient at successors, and take unions at limits; relation sim_delta is Sigma-in_(2*delta) definable.. Source1314-1333,1381-1391 and1530-1544; Ash1986, Ash-Knight2000 and Alvir-Rossegger2020.. Only the well-order building blocks and definability of block relations are allowed. The shuffle-sum construction, orbit definitions and two-round back-and-forth argument are not assumed.

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In this article we give a comprehensive Scott analysis of the class of linear orderings using three definitions of rank. Given a structure $\mathcal{A}$ the *(parameterless) Scott rank* $SR(\mathcal{A})$, is the least α, such that the automorphism orbits of all tuples in $\mathcal{A}$ are $\Sigma^{\text{in}}_\alpha$ definable; the *parameterized Scott rank* $pSR(\mathcal{A})$, is the least ordinal α such that there is a parameter $\bar{p} \in A^{<\omega}$ such that $SR((\mathcal{A}, \bar{p})) = \alpha$; and the *Scott sentence complexity* of $\mathcal{A}$, $SSC(\mathcal{A})$ is the least of the complexities Π^{in}_α, $\Sigma^{\text{in}}_\alpha$, $d\text{-}\Sigma^{\text{in}}_\alpha$ such that $\mathcal{A}$ has a Scott sentence of one of these complexities. While the latter is not an

0.1. **Preliminaries.** We assume that the reader is familiar with the basic techniques of Scott analysis and with linear orderings in general. For background we refer to Montalbán's upcoming book [4] and the standard reference by Ash and Knight [1]. Let us highlight the following lemma which we will use as basic tools in our proofs without reference.

Lemma 0.1 ([1, Lemma 15.8])**.** *Suppose $\mathcal{A}, \mathcal{B} \in \text{LO}$. Then $\mathcal{A} \leq_1 \mathcal{B}$ if and only if $\mathcal{A}$ is infinite or at least as large as $\mathcal{B}$. For $\alpha > 1$, $\mathcal{A} \leq_\alpha \mathcal{B}$ if and only if for every $1 \leq \beta < \alpha$ and every partition of $\mathcal{B}$ into intervals $\mathcal{B}_0, \ldots, \mathcal{B}_n$, with endpoints in $\mathcal{B}$, there is a partition of $\mathcal{A}$ into intervals $\mathcal{A}_1, \ldots, \mathcal{A}_n$ with endpoints in $\mathcal{A}$, such that $\mathcal{B}_i \leq_\beta \mathcal{A}_i$.*

Notice that this lemma particularly implies that $(\mathcal{A}, \bar{a}) \leq_\alpha (\mathcal{B}, \bar{b})$ if and only if $(-\infty, a_0) \leq_\alpha (-\infty, b_0)$, $(a_{|\bar{a}-1|}, \infty) \leq_\alpha (b_{|\bar{b}-1|}, \infty)$ and for all $0 \leq i < |\bar{a} - 1|$ $(a_i, a_{i+1}) \leq_\alpha (b_i, b_{i+1})$ [1, Lemma 15.7].

1. Possible Scott sentence complexities of linear orderings

We now exhibit possible Scott sentence complexities of linear orderings. Many of our examples are constructed by producing shuffle sums of linear orderings. Recall that given a set S of countable linear orderings, the *shuffle sum* $Sh(S)$ of S is obtained by partitioning $\mathbb{Q}$ into $|S|$ mutually dense sets K_i and replacing each point in K_i with a copy of S_i. Note that a linear order of the form $L \cdot \mathbb{Q}$ is a special case of a shuffle sum. We begin by proving a couple of lemmas about the back-and-forth relations of shuffle sums.

Theorem 1.1. *For any limit ordinal λ, there is a linear order of Scott sentence complexity $\Sigma^{\text{in}}_{\lambda+3}$.*

Lemma 1.2. *Let $\mathcal{A}$ be a shuffle sum and $\mathcal{B} = \mathcal{L} \cdot \mathbb{Q}$ for some ordering $\mathcal{L}$. If $\mathcal{A} \leq_\alpha \mathcal{B}$ then $\mathcal{A} + \mathcal{L} + \mathcal{B} \leq_{\alpha+2} \mathcal{A} + \mathcal{B}$.*

Proof. To show this, we describe a winning strategy for the $\exists$-player in the $\alpha + 2$-back-and-forth game. On the first turn, the $\exists$-player copies the selected tuple of the $\forall$-player using an isomorphism of the initial $\mathcal{A}$s and the final $\mathcal{B}$s. The only possible non-isomorphic interval is the one between the smallest element b in $\mathcal{B}$ and the largest element a in $\mathcal{A}$. In other words, we need to show that

$$\mathcal{A}_{>a} + \mathcal{L} + \mathcal{B}_{<b} \geq_{\alpha+1} \mathcal{A}_{>a} + \mathcal{B}_{<b}.$$

Because $\mathcal{A}$ and $\mathcal{B}$ are shuffle sums, $\mathcal{A}_{>a} \cong N + \mathcal{A}$ and $\mathcal{B}_{<b} \cong \mathcal{B} + K$ for some linear orderings N and K. This means that we only need to show that $\mathcal{A} + \mathcal{L} + \mathcal{B} \geq_{\alpha+1} \mathcal{A} + \mathcal{B}$ if we play isomorphically on N and K.

On the next turn, the $\forall$-player plays on $\mathcal{A} + \mathcal{L} + \mathcal{B}$, and the $\exists$-player responds on $\mathcal{A} + \mathcal{B}$ using an isomorphism between the initial $\mathcal{A}$s and an isomorphism of $\mathcal{L} + \mathcal{B}$ to a final segment of $\mathcal{B}$. Assume without loss of generality, that c is the smallest element in $\mathcal{L}$ played by the $\forall$-player in the distinguished copy of $\mathcal{L}$ within $\mathcal{A} + \mathcal{L} + \mathcal{B}$. The only possible non-isomorphic interval is between c and the largest element a selected within $\mathcal{A}$. In other words, we need to show that

$$\mathcal{A}_{>a} + \mathcal{L}_{<c} \leq_\alpha \mathcal{A}_{>a} + \mathcal{B} + \mathcal{L}_{<c}.$$

As $\mathcal{A}_{>a} \cong M + \mathcal{A}$ and the $\exists$-player can play the isomorphism between the copies of $\mathcal{L}_{<c}$ and M, it is sufficient to show that $\mathcal{A} \leq_\alpha \mathcal{A} + \mathcal{B}$. As $\mathcal{A}$ is a shuffle sum $\mathcal{A} \cong \mathcal{A} + \mathcal{A}$. Therefore, it is enough to show that $\mathcal{A} \leq_\alpha \mathcal{B}$, as desired. □

Corollary 1.3. *Let $\mathcal{A}$ be a shuffle sum and $\mathcal{B} = \mathcal{L} \cdot \mathbb{Q}$ for some ordering $\mathcal{L}$. If $\mathcal{A} + \mathcal{L} + \mathcal{B} \ncong \mathcal{A} + \mathcal{B}$ and $\mathcal{A} \leq_\alpha \mathcal{B}$, then $\mathcal{A} + \mathcal{L} + \mathcal{B}$ has no $\Pi^{\text{in}}_{\alpha+2}$ Scott sentence.*

Proof. We let $\mathcal{L}_1 := 1 + \mathbb{Z} + \omega^\lambda$ and $\mathcal{L}_2 := 1 + \mathbb{Z} + \omega^\lambda \cdot 2$. One important initial observation is that $\mathcal{L}_1 \leq_{\lambda+1} \mathcal{L}_2$ as $\omega^\lambda \leq_{\lambda+1} \omega^\lambda \cdot 2$. As alluded to above, we take $\mathcal{A} = \mathcal{L}_1 \cdot \mathbb{Q}$ and $\mathcal{B} = \mathcal{L}_2 \cdot \mathbb{Q}$. Finally we let $\mathcal{L} = \mathcal{A} + \mathcal{L}_2 + \mathcal{B}$ and claim that $\mathcal{L}$ has the desired complexity Let us begin with showing that $SR(\mathcal{A}) \leq \lambda$. We define several helping predicates from which the claim will easily follow.

(1) Let $pt(x)$ if x is in a block of size 1.
(2) Let $Z(x)$ say $\exists z < x\ pt(z) \wedge \forall y\ z < y < x \to \bigvee_n S^n(y) = x$. Note that $Z(x)$ holds if and only if x is inside one of the copies of $\mathbb{Z}$.
(3) Let $init(x)$ if $\exists z < x\ pt(z) \wedge (z, x) \cong \mathbb{Z}$. Note that $init(x)$ if and only if x is the first element in a copy of ω^λ.
(4) Let $\gamma(x)$ if $\exists z < x\ init(z) \wedge [z, x) \cong \gamma$ for any $\gamma < \omega^\lambda$. Note that this definition is of complexity lower than $\Sigma^{\text{in}}_\lambda$ as λ is a limit ordinal. It describes the points that are of rank γ within a copy of ω^λ.

Using these predicates one can define the automorphism orbits of any tuples in $\mathcal{A}$ using formulas of complexity less than $\Sigma^{\text{in}}_{\lambda+1}$.

All of these predicates have the same meaning within $\mathcal{B}$, however they leave points in the second copy of ω^λ undefined. But it is still the case that $SR(B) \leq \lambda+1$. To see this we need to define the additional relations

(1) Let

$$sec(x, z) \iff init(z) \wedge z < x \wedge \neg\exists y\ \ (z < y < x \wedge pt(y)) \wedge \bigwedge_{\delta<\lambda} \forall y < x\ y \not\sim_\delta x.$$

This Π^{in}_λ predicate holds only if z is the initial element in the second copy of ω^λ above the copy that z is initial in.
(2) Let $\gamma_2(x)$ if $\exists y, z\ sec(y, z) \wedge [y, x) \cong \gamma$ for any $\gamma \in \omega^\lambda$. Note that this definition is of complexity $\Sigma^{\text{in}}_{\lambda+1}$ as λ is a limit ordinal. It describes the points that are at the γ position within a second copy of ω^λ.

Using these relations and those defined above, one can define the automorphism orbits of tuples in $\mathcal{B}$.

In order to finish the proof of the claim, we appeal to Corollary 1.3 to show that there is no $\Pi^{\text{in}}_{\lambda+3}$ Scott sentence. This follows immediately as $\mathcal{A} \leq_{\lambda+1} \mathcal{B}$ is a direct consequence of $\mathcal{L}_1 \leq_{\lambda+1} \mathcal{L}_2$. □

References

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[4] A. Montalban. *Computable Structure Theory: Beyond the Arithmetic*. Draft,2021. `https://math.berkeley.edu/~antonio/CSTpart2_DRAFT.pdf`.